

Kidstuff

The Fedora 10 repository now includes Sugar, the learning software environment for the OLPC project. As yet, only a few activities are packaged in RPMs. I expect that more will be added as time passes. The Fedora project team hopes to get more people actively involved in the Sugar project by making the platform accessible to a wider population. I would strongly recommend that you try the *turtleart* activity, based on Logo. It is a colourful, fun way to learn programming.

RPMFusion

In addition to the Fedora 10 release, the availability of RPMFusion repositories has been extremely valuable. The confusion between whether to use Livna or FreshRPMs is over. The migration has been transparent for all those who were using either of these two repositories and conflicts between the packages have been ironed out.

Pre-upgrade

The pre-upgrade utility has become very useful with Fedora10. The idea is that it will analyse the packages that are installed and download the required upgrades while you continue working. The utility will also ensure that dependencies are not destroyed for the packages that have been installed from alternate repositories. This is the first time I did not have to do anything to ensure that the multimedia functions worked for the various formats, even after the upgrade. The steps involved are as follows:

```
# yum install preupgrade
# preupgrade
```

On my system, it downloaded 1.8 GB of packages in 24 hours. If you stop in the middle, it restarts from where it left off. Once the packages are downloaded, reboot the system and it will install the upgrade.

The upgrade failed once. It needed about 1.5 GB of free space. I could boot normally, create the desired space and run *preupgrade* again. This time, the upgrade was uneventful.

This step took a little over two hours. So, the effective downtime was two hours. A fresh install will be faster, but will need all the settings to be redone and the additional packages to be downloaded.

The migration to Fedora 10 was effortless. Everything worked fine after the upgrade, including MPlayer, VLC, and MP3 playback. After that, I used *yum update* to upgrade the multimedia packages.

Using update to upgrade

It is possible to update Fedora 10 with virtually zero downtime using an unsupported process. I had first come across www.ioncannon.net/linux/68/upgrading-from-fc6-to-fedora7-with-yum last year and used this technique for upgrading from Fedora 7 to 8 and then from 8 to 9. On both occasions, I had a few problems with some multimedia packages. This time, the process was remarkably smooth thanks to the availability of

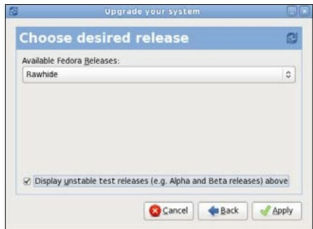


Figure 1: Select an available update, observe and reboot when ready



Figure 2: The KDE 4.2 desktop with an instance of Miro running

RPMFusion repositories as well. The steps involved are:

1. Download the following packages from the Fedora 10 repository:
`fedora-release-10-1.noarch.rpm`
`fedora-release-notes-10.0-1.noarch.rpm`
`yum-3.2.20-3.fc10.noarch.rpm`
2. Use *rpm -U* to update the above three packages
3. Clean the existing repositories using *yum clean all*
4. Finally, run *yum update*

The fourth step will take a very long time to first download the packages. On my parents' system, it needed to download 1.2 GB and took about 18 hours. The update went on in the background for over two hours. As libraries and packages get replaced, some applications may present a problem, but I did not face any. I wasn't doing anything serious—just playing music and browsing.

If an installation DVD is available (like the one bundled with this month's *LFY*), copy the RPMs into the `/var/cache/yum/fedora/packages` after Step 3, and the update will download only the missing or updated packages.

I find this method very useful for small networks. The cache directory can be shared over the network and the *keepcache* option can be set to 1 in *yum.conf*.